

1 Illustrative Examples

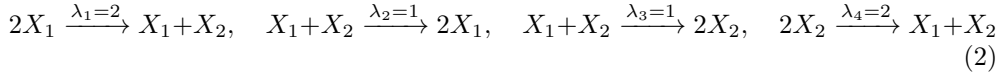
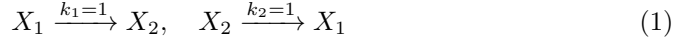
We present examples in two groups. The first group consists of two analytically tractable cases where the inference problem is provably non-identifiable; these require no numerical computation and serve to illustrate the theoretical conditions of ???. The second group consists of two computational examples where all identifiability conditions are satisfied. For the numerical examples, we compare our Bayesian MCMC approach against the sparse-learning-CRN method of [?], which recovers propensity functions by solving an ℓ_1 -regularized least-squares problem (LASSO) over a polynomial basis using the FISTA algorithm. (All code and data are available at [BayesCRNInference](#).)

Structural Non-Identifiability: Two Illustrative Cases

The following two cases both satisfy ??? via a non-negative conservation law. In each case the violation of condition (??) persists for all trajectory lengths: no amount of data can resolve the ambiguity. We do not report MCMC results for these cases; the non-identifiability is established analytically, and running the sampler would merely confirm that the posterior is diffuse over an equivalence class of parameter vectors.

First case: stoichiometric conservation law.

Following [?], consider two systems on species (X_1, X_2) sharing the conservation law $X_1 + X_2 = 2$:



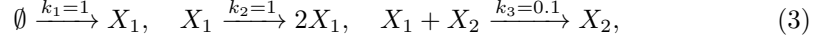
The conservation law confines the state space to $\{(2, 0), (1, 1), (0, 2)\}$. Table 1 shows that both systems produce identical propensities at every accessible state, so condition (??) is violated for every member of the interpolating family $\theta(\alpha) = \alpha \theta_{\text{Sys.1}} + (1 - \alpha) \theta_{\text{Sys.2}}$. The conservation vector $c = (1, 1)^\top$ exists and the channel $\xi_\ell = (-1, +1)$ contains the unimolecular reaction $X_1 \rightarrow X_2$ ($\|\nu\|_1 = 1$), satisfying condition (2) of ???. Structural non-identifiability follows regardless of trajectory length.

State (X_1, X_2)	$(\Delta X_1, \Delta X_2)$	Sys. (1) propensity	Sys. (2) propensity
(2, 0)	(-1, +1)	$k_1 \cdot 2 = 2$	$\lambda_1 \cdot \frac{2(2-1)}{2} = 2$
(1, 1)	(-1, +1)	$k_1 \cdot 1 = 1$	$\lambda_3 \cdot 1 \cdot 1 = 1$
(1, 1)	(+1, -1)	$k_2 \cdot 1 = 1$	$\lambda_2 \cdot 1 \cdot 1 = 1$
(0, 2)	(+1, -1)	$k_2 \cdot 2 = 2$	$\lambda_4 \cdot \frac{2(2-1)}{2} = 2$

Table 1 Propensities for Systems (1) and (2) with $X_1 + X_2 = 2$. The two systems are indistinguishable at every accessible state.

Second case: catalyst species.

Consider the two-species system

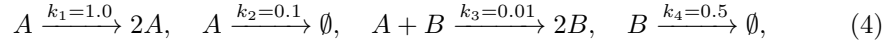


with propensities $a_1 = k_1$, $a_2 = k_2 X_1(X_1 - 1)/2$, $a_3 = k_3 X_1 X_2$. Since no reaction in (3) produces or degrades X_2 , the species acts as a permanent catalyst: $X_2(t) = c_2$ for all t . This constitutes a conservation law with $c = (0, 1)^\top$, and ?? applies again. In the channel $\xi_\ell = (-1, 0)$, the candidate reaction $X_1 \rightarrow \emptyset$ is unimolecular ($\|\nu\|_1 = 1$), satisfying condition (2). The propensities $k X_1$ (for $X_1 \rightarrow \emptyset$) and $k' c_2 X_1$ (for $X_1 + X_2 \rightarrow X_2$) are proportional for all visited states, so the full-rank condition (??) fails and any pair $(\tilde{k}_1, \tilde{k}_3)$ with $\tilde{k}_1 + \tilde{k}_3 c_2 = k_3 c_2$ is equally consistent with the data. In contrast to the first case, where non-identifiability arises from the topology of the state space, here it arises from collinearity of the propensity functions - but both are structural, persisting in every trajectory generated by these networks.

Numerical Example 1: Predator-Prey System

Both numerical examples satisfy all identifiability conditions of ?. Throughout, Bayesian MCMC is run with a spike-and-slab prior ($\pi = 0.75$, $\alpha = 2$, $\beta = 0.25$), 500,000 iterations, 50,000 burn-in, thinning by 100, and rates initialized from $U[0.001, 2.0]$, with covariance updated adaptively every 10 iterates. For comparison, the LASSO method of [?] is applied at regularization strength $\lambda = 0.01$ unless stated otherwise.

We consider the stochastic Lotka-Volterra system with prey A and predator B :



with mass-action propensities $\lambda_1 = k_1 A$, $\lambda_2 = k_2 A$, $\lambda_3 = k_3 AB$, $\lambda_4 = k_4 B$. The deterministic equilibrium is $(A^*, B^*) = (50, 90)$. Starting from $(A, B) = (100, 50)$, the system is simulated over $T = 40$, yielding 5,805 events across four stoichiometric channels. The large initial displacement from equilibrium drives pronounced oscillatory dynamics, with stochastic prey extinction occurring near $t = 39.8$.

We search over all unimolecular and bimolecular candidate reactions for the two species system under the assumption of mass action kinetics. The rank condition (??) is satisfied for each active channel. Figure 1 shows the simulated trajectory. Table 2 shows all candidate reactions considered for each stoichiometric channel, together with the posterior results. Our Bayesian approach correctly identifies all four stoichiometric channels, with posterior means within 2% of the true rates and $\text{Pr}(\text{off}) = 0$ for all active reactions. All inactive candidate reactions receive $\text{Pr}(\text{off}) \approx 1$.

The LASSO was applied with a polynomial basis of degree at most 2, $\{1, A, B, A^2, AB, B^2\}$ at regularization strength $\lambda = 0.01$. This method performs well on the channels $(+1, 0)$, $(-1, +1)$, and $(0, -1)$, but fails on channel $(-1, 0)$ corresponding to $A \rightarrow \emptyset$. Rather than recovering a propensity proportional to A , it returns $0.018 A + 0.036 B$, assigning the larger coefficient to B . We note that this would be

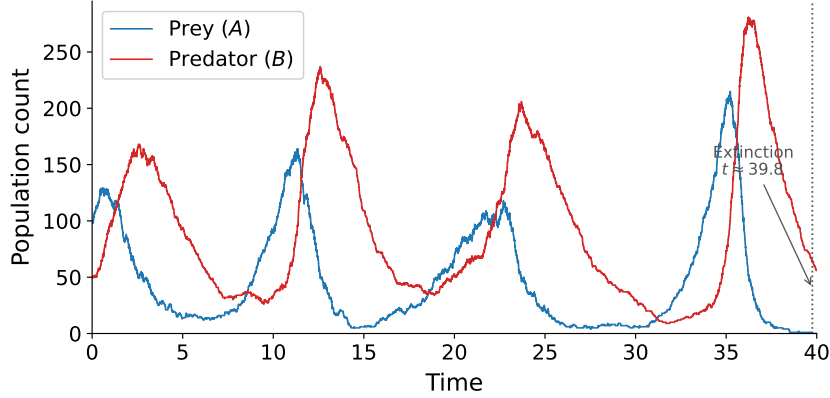


Fig. 1 Stochastic trajectory of the predator-prey system (4) simulated over $T = 40$ from initial state $(A, B) = (100, 50)$. Prey (A , blue) and predator (B , red) undergo pronounced oscillations driven by the large displacement from the deterministic equilibrium $(A^*, B^*) = (50, 90)$. Prey extinction occurs near $t \approx 39.8$ (dotted line), yielding 5,805 observed events across four stoichiometric channels.

mechanistically impossible under the assumption of mass-action kinetics. However, this failure has an understandable structural cause. Near the deterministic equilibrium $B^* = 90 > A^* = 50$, values of B are systematically larger than values of A , so assigning weight to B achieves a similar propensity contribution at lower ℓ_1 cost. The oscillation-induced correlation between A and B further compounds this, and the low event count (190 jumps) provides insufficient signal to overcome it. Because the Bayesian approach works directly in the space of reaction rates under mass-action constraints (rather than fitting unconstrained polynomial coefficients) it is not susceptible to this scale imbalance.

We note that [?] also demonstrated their method on a two-species predator-prey system with the same four reaction types, using 100 independent trajectories of length $T = 10$ and a smaller regularization parameter $\lambda = 0.0001$; the substantially larger data volume (equivalent to 1,000 total time units versus our single trajectory of length $T = 40$) and the absence of extinction events likely reduce the oscillation-induced correlation between A and B that causes the failure on channel $(-1, 0)$. Our comparison thus reflects a regime where data are limited and the trajectory is far from stationarity, which is precisely where the structural constraints of the Bayesian formulation provide the greatest advantage.

Numerical Example 2: Randomly Generated Reaction Network

Using the same MCMC and LASSO settings as in the previous example (polynomial order 2), we assess performance at scale with a four-species system (A, B, X, Y) , enumerating all unimolecular and bimolecular reactions to form a candidate space of 130 stoichiometric channels. Twenty reactions are selected at random (with 4 channels chosen to contain 2 active reactions each), and rate parameters are independently drawn from $\text{Gamma}(2.6, 0.4)$. A single trajectory is simulated from $(A, B, X, Y) = (5, 5, 5, 5)$

Table 2 Predator-prey inference at $T = 40$ (5,805 events total). All candidate mass-action reactions for each stoichiometric channel ξ_ℓ are shown. k^* : true rate (bold if active). $\hat{k}$: posterior mean (95% credible interval). Pr(off): spike-and-slab exclusion probability. LASSO propensity at $\lambda = 0.01$. Entries marked “—” have Pr(off) > 0.99 : the posterior is overwhelmingly concentrated on the spike component (rate = 0), so the credible interval collapses to a point at zero and is not reported. † Mechanistically impossible under mass-action kinetics.

ξ_ℓ	Candidate reaction	k^*	Events	$\hat{k}$ (95% CI)	Pr(off)	LASSO
(+1, 0)	$\emptyset \rightarrow A$	0	1933	—	0.995	
	$A \rightarrow 2A$	1.0000		1.0024 (0.9587, 1.0480)	0.000	0.969 A
	$B \rightarrow A + B$	0		—	1.000	
(-1, 0)	$A \rightarrow \emptyset$	0.1000	190	0.0995 (0.0859, 0.1140)	0.000	0.018 A + 0.036 B †
	$2A \rightarrow A$	0		—	1.000	
	$A + B \rightarrow B$	0		—	1.000	
(-1, +1)	$A \rightarrow B$	0	1843	—	1.000	
	$2A \rightarrow A + B$	0		—	1.000	
	$A + B \rightarrow 2B$	0.0100		0.0101 (0.0096, 0.0105)	0.000	0.010 AB
(0, -1)	$B \rightarrow \emptyset$	0.5000	1837	0.4918 (0.4705, 0.5144)	0.000	0.497 B
	$2B \rightarrow B$	0		—	1.000	
	$A + B \rightarrow A$	0		—	1.000	

over $T \in \{100, 200, 400\}$; all species remain at low copy numbers throughout (maximum count 20), as shown in Figure 2, making structural identification from short trajectories particularly challenging.

Results in Tables 3–7 show that the spike-and-slab prior achieves exact structural recovery at all three time horizons: every inactive reaction receives Pr(off) = 1.000 and every active reaction Pr(off) = 0.000, with posterior widths narrowing as T increases. This includes the four channels containing two simultaneously active reactions, where both reactions are correctly identified in each case. Even the $-2A$ channel, observed only 60 times at $T = 100$, is correctly identified with Pr(off) = 0 and a posterior mean within 3% of the true rate (Table 3).

At $T = 100$, the Bayesian method achieves lower L^2 propensity error than LASSO on every channel (errors 0.008–0.168 vs. 0.143–0.991; Table 6). Notably, LASSO errors improve little or worsen from $T = 200$ to $T = 400$ on several channels (e.g., $-2A, +1X, +1Y$: 0.754 $\rightarrow$ 0.851 $\rightarrow$ 0.993), whereas Bayesian errors consistently shrink with more data. A sweep over $\lambda \in \{0.001, 0.005, 0.01, 0.05, 0.1, 0.5\}$ confirms that no single regularization strength minimizes error across all channels simultaneously: weak-reaction channels require small λ to avoid over-shrinkage, while noisy channels require large λ for suppression (Table 7). The Bayesian framework resolves this automatically through the spike-and-slab prior, adapting to each channel’s evidence without global tuning.

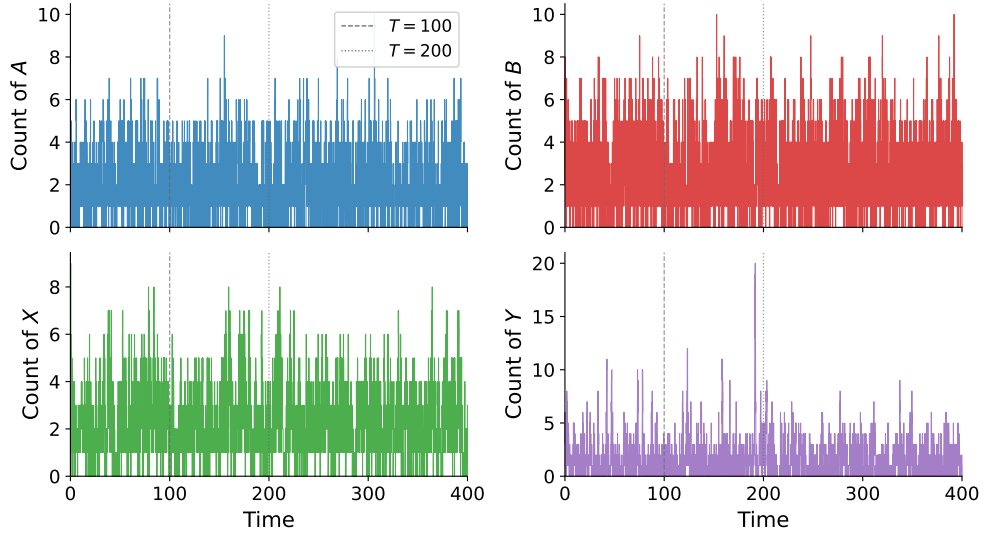


Fig. 2 Stochastic trajectory of the randomly generated four-species system (A, B, X, Y) , simulated from $(5, 5, 5, 5)$ over $T = 400$. Each panel shows one species; the dashed and dotted vertical lines mark the $T = 100$ and $T = 200$ observation horizons used in the inference experiments. All species remain at low copy numbers throughout, making structural identification from short trajectories challenging.

Table 3 Bayesian spike-and-slab posterior estimates, $T = 100$. Stoichiometric change ξ_ℓ and observed event count (Obs) are shown once per channel. True rate k^* is bold for active reactions. $\hat{k}$ and 95% credible interval are combined. Candidates with $\text{Pr}(\text{off}) > 0.99$ are omitted.

Change	Obs	Reaction	True k^*	$\hat{k}$ (95% CI)	Pr(off)
$-2A$	60	$2A \rightarrow \emptyset$	0.4528	0.4416 [0.341, 0.559]	0.0000
$-2A, +1X, +1Y$	129	$2A \rightarrow X + Y$	1.0206	0.9363 [0.783, 1.105]	0.0000
$-2A, +1B$	230	$2A \rightarrow B$	1.5713	1.6576 [1.451, 1.869]	0.0000
$-2B, +1X$	615	$2B \rightarrow X$	1.9789	2.0576 [1.904, 2.221]	0.0000
$-1X, +1Y$	138	$X + Y \rightarrow 2Y$	0.4887	0.5338 [0.446, 0.625]	0.0000
$-1Y$	310	$A + Y \rightarrow A$	0.5897	0.4821 [0.344, 0.635]	0.0000
		$X + Y \rightarrow X$	0.7676	0.7801 [0.638, 0.925]	0.0000
$+1Y$	184	$A \rightarrow A + Y$	0.2803	0.2617 [0.164, 0.373]	0.0002
		$Y \rightarrow 2Y$	1.0673	0.9431 [0.772, 1.125]	0.0000
$+1X, -1Y$	136	$A + Y \rightarrow A + X$	0.6456	0.6117 [0.514, 0.716]	0.0000
$+1B, -1X$	427	$2X \rightarrow B + X$	1.2593	1.2039 [1.003, 1.413]	0.0000
		$A + X \rightarrow A + B$	0.4167	0.3291 [0.226, 0.443]	0.0000
$+1B, -1X, +1Y$	317	$X \rightarrow B + Y$	1.2783	1.3700 [1.227, 1.526]	0.0000
$+1B, -1Y$	326	$A + Y \rightarrow A + B$	0.4221	0.3658 [0.234, 0.512]	0.0000
		$X + Y \rightarrow B + X$	0.9061	0.9419 [0.793, 1.095]	0.0000
$+1B$	378	$A \rightarrow A + B$	1.0464	1.2137 [0.948, 1.488]	0.0000
		$X \rightarrow B + X$	0.7167	0.7369 [0.553, 0.937]	0.0000
$+2B$	64	$\emptyset \rightarrow 2B$	0.6782	0.6586 [0.511, 0.824]	0.0000
$+1A, -1B$	580	$B \rightarrow A$	2.2031	2.2523 [2.072, 2.445]	0.0007
$+1A$	255	$A \rightarrow 2A$	1.5168	1.4763 [1.302, 1.661]	0.0000

Table 4 Bayesian spike-and-slab posterior estimates, $T = 200$. Stoichiometric change ξ_ℓ and observed event count (Obs) are shown once per channel. True rate k^* is bold for active reactions. $\hat{k}$ and 95% credible interval are combined. Candidates with $\text{Pr}(\text{off}) > 0.99$ are omitted.

Change	Obs	Reaction	True k^*	$\hat{k}$ (95% CI)	Pr(off)
$-2A$	115	$2A \rightarrow \emptyset$	0.4528	0.4230 [0.351, 0.503]	0.0000
$-2A, +1X, +1Y$	263	$2A \rightarrow X + Y$	1.0206	0.9590 [0.849, 1.078]	0.0000
$-2A, +1B$	452	$2A \rightarrow B$	1.5713	1.6443 [1.499, 1.797]	0.0000
$-2B, +1X$	1185	$2B \rightarrow X$	1.9789	2.0292 [1.914, 2.147]	0.0000
$-1X, +1Y$	252	$X + Y \rightarrow 2Y$	0.4887	0.5010 [0.444, 0.562]	0.0000
$-1Y$	626	$A + Y \rightarrow A$	0.5897	0.5647 [0.455, 0.685]	0.0000
		$X + Y \rightarrow X$	0.7676	0.7817 [0.677, 0.891]	0.0000
$+1Y$	398	$A \rightarrow A + Y$	0.2803	0.2773 [0.204, 0.358]	0.0000
		$Y \rightarrow 2Y$	1.0673	1.0147 [0.890, 1.146]	0.0000
$+1X, -1Y$	264	$A + Y \rightarrow A + X$	0.6456	0.6411 [0.567, 0.718]	0.0000
$+1B, -1X$	860	$2X \rightarrow B + X$	1.2593	1.1867 [1.044, 1.337]	0.0000
		$A + X \rightarrow A + B$	0.4167	0.3812 [0.304, 0.461]	0.0000
$+1B, -1X, +1Y$	603	$X \rightarrow B + Y$	1.2783	1.3327 [1.228, 1.440]	0.0000
$+1B, -1Y$	629	$A + Y \rightarrow A + B$	0.4221	0.3698 [0.273, 0.477]	0.0000
		$X + Y \rightarrow B + X$	0.9061	0.9452 [0.837, 1.054]	0.0000
$+1B$	713	$A \rightarrow A + B$	1.0464	1.0272 [0.851, 1.204]	0.0000
		$X \rightarrow B + X$	0.7167	0.8016 [0.675, 0.944]	0.0000
$+2B$	127	$\emptyset \rightarrow 2B$	0.6782	0.6461 [0.541, 0.761]	0.0000
$+1A, -1B$	1143	$B \rightarrow A$	2.2031	2.2706 [2.137, 2.403]	0.0000
$+1A$	512	$A \rightarrow 2A$	1.5168	1.4980 [1.371, 1.627]	0.0000

Table 5 Bayesian spike-and-slab posterior estimates, $T = 400$. Stoichiometric change ξ_ℓ and observed event count (Obs) are shown once per channel. True rate k^* is bold for active reactions. $\hat{k}$ and 95% credible interval are combined. Candidates with $\text{Pr}(\text{off}) > 0.99$ are omitted.

Change	Obs	Reaction	True k^*	$\hat{k}$ (95% CI)	Pr(off)
$-2A$	232	$2A \rightarrow \emptyset$	0.4528	0.4409 [0.386, 0.499]	0.0000
$-2A, +1X, +1Y$	533	$2A \rightarrow X + Y$	1.0206	1.0090 [0.924, 1.096]	0.0000
$-2A, +1B$	893	$2A \rightarrow B$	1.5713	1.6866 [1.579, 1.800]	0.0000
$-2B, +1X$	2346	$2B \rightarrow X$	1.9789	2.0054 [1.923, 2.088]	0.0000
$-1X, +1Y$	482	$X + Y \rightarrow 2Y$	0.4887	0.4832 [0.441, 0.527]	0.0000
$-1Y$	1232	$A + Y \rightarrow A$	0.5897	0.5656 [0.481, 0.655]	0.0000
		$X + Y \rightarrow X$	0.7676	0.7964 [0.719, 0.875]	0.0000
$+1Y$	766	$A \rightarrow A + Y$	0.2803	0.2703 [0.217, 0.328]	0.0000
		$Y \rightarrow 2Y$	1.0673	1.0607 [0.968, 1.158]	0.0000
$+1X, -1Y$	515	$A + Y \rightarrow A + X$	0.6456	0.6655 [0.609, 0.723]	0.0000
$+1B, -1X$	1717	$2X \rightarrow B + X$	1.2593	1.2025 [1.100, 1.311]	0.0000
		$A + X \rightarrow A + B$	0.4167	0.4013 [0.346, 0.459]	0.0000
$+1B, -1X, +1Y$	1197	$X \rightarrow B + Y$	1.2783	1.3182 [1.243, 1.394]	0.0000
$+1B, -1Y$	1236	$A + Y \rightarrow A + B$	0.4221	0.3912 [0.315, 0.474]	0.0000
		$X + Y \rightarrow B + X$	0.9061	0.9360 [0.861, 1.019]	0.0000
$+1B$	1400	$A \rightarrow A + B$	1.0464	1.0716 [0.947, 1.199]	0.0000
		$X \rightarrow B + X$	0.7167	0.7584 [0.667, 0.851]	0.0000
$+2B$	252	$\emptyset \rightarrow 2B$	0.6782	0.6343 [0.558, 0.713]	0.0000
$+1A, -1B$	2260	$B \rightarrow A$	2.2031	2.2325 [2.139, 2.326]	0.0000
$+1A$	1054	$A \rightarrow 2A$	1.5168	1.5844 [1.491, 1.682]	0.0000

Table 6 Polynomial-space L^2 propensity error, $\|\hat{c} - c_{\text{true}}\|_2$, for each stoichiometric channel across trajectory lengths T . **Bold**: lower error for each (channel, T) pair. LASSO uses fixed $\lambda = 0.01$.

Change	$T = 100$		$T = 200$		$T = 400$	
	Bayes	LASSO	Bayes	LASSO	Bayes	LASSO
$-2A$	0.008	0.487	0.021	0.374	0.008	0.317
$-2A, +1X, +1Y$	0.060	0.754	0.044	0.851	0.008	0.993
$-2A, +1B$	0.061	0.962	0.052	0.879	0.082	0.938
$-2B, +1X$	0.056	0.843	0.036	1.005	0.019	1.009
$-1X, +1Y$	0.045	0.226	0.012	0.292	0.006	0.320
$-1Y$	0.108	0.632	0.029	0.411	0.038	0.174
$+1Y$	0.126	0.484	0.053	0.451	0.012	0.178
$+1X, -1Y$	0.034	0.143	0.005	0.188	0.020	0.168
$+1B, -1X$	0.096	0.832	0.062	0.816	0.043	0.686
$+1B, -1X, +1Y$	0.092	0.360	0.054	0.308	0.040	0.182
$+1B, -1Y$	0.067	0.209	0.065	0.160	0.043	0.153
$+1B$	0.168	0.713	0.087	0.342	0.049	0.283
$+2B$	0.020	0.797	0.032	0.163	0.044	0.519
$+1A, -1B$	0.049	0.991	0.067	0.411	0.029	0.195
$+1A$	0.040	0.408	0.019	0.262	0.068	0.152

Table 7 Polynomial-space L^2 error for sparse-learning-CRN at $T=200$ across regularisation strengths λ . Obs = observed event count in the $T=200$ trajectory. **Bold**: minimum L^2 per channel. No single λ minimises error across all channels simultaneously, illustrating the fundamental limitation of global regularisation.

Change	Obs	$\lambda = 0.001$	$\lambda = 0.005$	$\lambda = 0.01$	$\lambda = 0.05$	$\lambda = 0.1$	$\lambda = 0.5$
$-2A$	115	1.087	0.431	0.374	0.294	0.251	0.245
$-2A, +1X, +1Y$	263	1.477	0.919	0.851	0.632	0.570	0.556
$-2A, +1B$	452	0.919	0.881	0.879	0.866	0.832	0.833
$-2B, +1X$	1185	1.215	1.044	1.005	0.693	0.953	1.032
$-1X, +1Y$	252	1.867	0.763	0.292	0.083	0.029	0.073
$-1Y$	626	0.750	0.575	0.411	0.078	0.057	0.201
$+1Y$	398	0.540	0.378	0.451	0.481	0.648	1.124
$+1X, -1Y$	264	0.388	0.234	0.188	0.078	0.038	0.120
$+1B, -1X$	860	1.153	0.993	0.816	0.695	0.668	0.690
$+1B, -1X, +1Y$	603	0.594	0.465	0.308	0.155	0.322	1.315
$+1B, -1Y$	629	0.176	0.170	0.160	0.083	0.084	0.202
$+1B$	713	0.452	0.393	0.342	0.493	0.568	1.309
$+2B$	127	0.456	0.219	0.163	0.310	0.361	0.687
$+1A, -1B$	1143	0.375	0.236	0.411	0.294	0.637	2.249
$+1A$	512	0.492	0.302	0.262	0.542	0.786	1.547